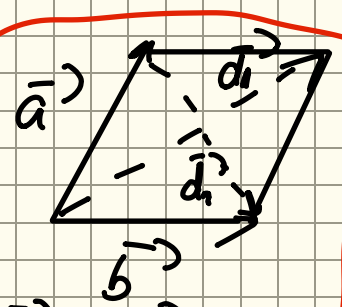
1.7) $\vec{a} = 5\vec{p} - 3\vec{q}$, $\vec{b} = \vec{p} + 2\vec{q}$

$\|\vec{p}\| = 3$ $\quad \alpha = \sphericalangle(\vec{p}, \vec{q}) = \frac{\pi}{3}$

$|\vec{q}| = 2$



$\vec{a} + \vec{b} = \vec{d_1}$

$\vec{a} - \vec{b} = \vec{d_2}$

a) $\vec{a} + \vec{b} = 6\vec{p} - \vec{q}$

$\vec{a} - \vec{b} = 4\vec{p} - 5\vec{q}$

$\vec{p} \cdot \vec{q} = |\vec{p}| \cdot |\vec{q}| \cdot \cos\alpha = 3 \cdot 2 \cdot \frac{1}{2} = 3$

$|\vec{a} + \vec{b}|^2 = (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = (6\vec{p} - \vec{q})(6\vec{p} - \vec{q})$

$= 36|p|^2 - 12\vec{p}\cdot\vec{q} + |q|^2 = 36 \cdot 9 - 12 \cdot 3 + 4$

$= 288 + 4 = 292 \Rightarrow |\vec{a} + \vec{b}| = \sqrt{292}$

$|\vec{a} - \vec{b}|^2 = 16|p|^2 - 40\vec{p}\cdot\vec{q} + 25|q|^2 = 16 \cdot 9 - 120 + 100$

$= 144 - 20 = 124$

$\Rightarrow |\vec{a} - \vec{b}| = \sqrt{124}$

$|\vec{c}|^2 = \vec{c} \cdot \vec{c}$

b) $\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos\theta$

$|\vec{a}|^2 = \vec{a} \cdot \vec{a} = 25|\vec{p}|^2 - 30\vec{p}\cdot\vec{q} + 9|\vec{q}|^2$

$= 225 - 90 + 36 = 135 + 36 = 171 \Rightarrow |\vec{a}| = \sqrt{171}$

$|\vec{b}|^2 = \vec{b} \cdot \vec{b} = |\vec{p}|^2 + 4\vec{p}\cdot\vec{q} + 4|\vec{q}|^2 = 9 + 12 + 64$

$= 85$

$\Rightarrow |\vec{b}| = \sqrt{85}$, $\quad \theta = \sphericalangle(d_1, d_2)$

$\Rightarrow \cos\theta = \vec{a}$

$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = |\vec{a} + \vec{b}| \cdot |\vec{a} - \vec{b}| \cdot \cos\theta$

$(\vec{a} + \vec{b})(\vec{a} - \vec{b}) = (6\vec{p} - \vec{q})(4\vec{p} - 5\vec{q}) =$

$= 24|\vec{p}|^2 - 30\vec{p}\cdot\vec{q} - 4\vec{p}\cdot\vec{q} + 5\cdot|q|^2$

$= 24\cdot 9 - 36\cdot 3 + 5\cdot 4 = 216 - 108 + 20$

$= 128$

$\Rightarrow \cos\theta = \frac{128}{\sqrt{171}\cdot\sqrt{292}} \Rightarrow \theta = \arccos\left(\frac{128}{\sqrt{171}\cdot\sqrt{292}}\right)$

c) $|\vec{a}\times\vec{b}| = |(5\vec{p}-3\vec{q})\times(\vec{p}+2\vec{q})|$

$= |5\vec{p}\times\vec{p} + 5\cdot 2(\vec{p}\times\vec{q}) - 3(\vec{q}\times\vec{p}) - 6(\vec{q}\times\vec{q})|$

$= 10\cdot 3\sqrt{3} + 3\cdot 3\sqrt{3} = 39\sqrt{3}$

$\vec{p}\times\vec{q} = |\vec{p}|\cdot|\vec{q}|\cdot\sin\frac{\pi}{3} = 3\cdot 2\cdot\frac{\sqrt{3}}{2} = 3\sqrt{3}$

1. 8) $\vec{a} = 12\vec{i} + 3\vec{j} - 4\vec{k}$

$\vec{b} = 3\vec{i} + 4\vec{j} + 12\vec{k}$

$\vec{a}\perp\vec{b} \Leftrightarrow \vec{a}\cdot\vec{b} = 0 \Rightarrow \vec{a}\cdot\vec{b} = 36 + 12 - 48 = 0$

$\Rightarrow \vec{a}\perp\vec{b}$ (1)

$|\vec{a}| = \sqrt{144 + 9 + 16} = \sqrt{169} = 13$

$|\vec{b}| = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$

$\Rightarrow |\vec{a}| = |\vec{b}|$ (2)

$\vec{a}\cdot\vec{b} = 0$, $|\vec{a}| = |\vec{b}| = L \Rightarrow \vec{c} = \pm\frac{\vec{a}\times\vec{b}}{L}$

From (1) and (2) $\Rightarrow \vec{a}$ and $\vec{b}$ are two edges of a cube

$\vec{c} = \pm\frac{(\vec{a}\times\vec{b})}{L} = \pm\frac{52\vec{i} - 156\vec{j} + 39\vec{k}}{13} = \pm(4\vec{i} - 12\vec{j} + 3\vec{k})$

$\vec{a}\times\vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 12 & 3 & -4 \\ 3 & 4 & 12 \end{vmatrix} = 52\vec{i} - 156\vec{j} + 39\vec{k}$

1. a) $\vec{v} = 2\vec{i} - 3\vec{j} - 5\vec{k}$

$\vec{v_1} = 3\vec{i} + \vec{k}$

$\vec{v_2} = -\vec{j} - \vec{k}$

$\vec{v_3} = -\vec{i} + 3\vec{k}$

$\vec{v} = \alpha\vec{v_1} + \beta\vec{v_2} + \gamma\vec{v_3}$

$\vec{v} = 3\alpha\vec{i} + \alpha\vec{k} - \beta\vec{j} - \beta\vec{k} - \gamma\vec{i} + 3\gamma\vec{k}$

$\vec{v} = \vec{i}(3\alpha - \gamma) - \beta\vec{j} + \vec{k}(\alpha - \beta + 3\gamma) = 2\vec{i} - 3\vec{j} - 5\vec{k}$

$$\begin{cases} 3\alpha - \gamma = 2 \\ -\beta = -3 \\ \alpha - \beta + 3\gamma = -5 \end{cases} \Rightarrow \begin{cases} \beta = 3 \\ 3\alpha - \gamma = 2 \\ \alpha + 3\gamma = -2 \end{cases} \Leftrightarrow \begin{cases} 9\alpha - 3\gamma = 6 \\ \alpha + 3\gamma = -2 \end{cases}$$

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$10\alpha = 4 \Rightarrow \alpha = \frac{2}{5}$

$\gamma = -\frac{4}{5}$

$\Rightarrow \vec{v} = \frac{2}{5}\cdot\vec{v_1} + 3\vec{v_2} - \frac{4}{5}\vec{v_3}$

1. 10) $\vec{v} = \vec{i} + \vec{j} + 2\vec{k}$

$\vec{a} = 2\vec{i} + \vec{j}$

$\vec{b} = -\vec{i} + 3\vec{k}$

$\vec{c} = -\vec{i} + \vec{j} - \vec{k}$

$\vec{v} = \alpha\vec{a} + \beta\vec{b} + \gamma\vec{c}$

$\vec{v} = 2\alpha\vec{i} + \alpha\vec{j} - \beta\vec{i} + 3\beta\vec{k} - \gamma\vec{i} + \gamma\vec{j} - \gamma\vec{k}$

$\vec{v} = \vec{i}(2\alpha - \beta - \gamma) + \vec{j}(\alpha + \gamma) + \vec{k}(3\beta - \gamma) = \vec{i} + \vec{j} + 2\vec{k}$

$$\begin{cases} 2\alpha - \beta - \gamma = 1 \\ \alpha + \gamma = 1 \\ 3\beta - \gamma = 2 \end{cases}$$

$$[A|B] = \left[\begin{array}{ccc|c} 2 & -1 & -1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 3 & -1 & 2 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 2 & -1 & -1 & 1 \\ 0 & 3 & -1 & 2 \end{array}\right] \overset{L_2 - 2L_1}{\sim} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & -1 & -3 & -1 \\ 0 & 3 & -1 & 2 \end{array}\right] \overset{L_3 + 3L_2}{\sim}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & -1 & -3 & -1 \\ 0 & 0 & -10 & -1 \end{array}\right] \Leftrightarrow \begin{array}{l} \alpha + \gamma = 1 \\ -\beta - 3\gamma = -1 \\ \gamma = \frac{1}{10}, \alpha = \frac{9}{10}, \beta = \frac{7}{10} \end{array}$$

$$\Rightarrow \vec{v} = \frac{9\vec{a} + 7\vec{b} + \vec{c}}{10}$$

1.11) $\vec{a} = 2\vec{i} + \vec{j}$

$\vec{b} = -\vec{i} + 3\vec{k}$

$\vec{c} = -\vec{i} + \vec{j} - \vec{k}$

$\vec{a}, \vec{b}, \vec{c}$ formează bază $\Rightarrow$

$$\begin{vmatrix} 2 & -1 & -1 \\ 1 & 0 & 1 \\ 0 & 3 & -1 \end{vmatrix} \neq 0 = 2\cdot(-3) - 1 \cdot (-3) = -10 \neq 0$$

$$\vec{d} = 2\vec{i} + \vec{j} + \vec{k} = \alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = 2\alpha\vec{i} + \alpha\vec{j} - \beta\vec{i} + 3\beta\vec{k} - \gamma\vec{i} + \gamma\vec{j} - \gamma\vec{k} = \vec{i}(2\alpha - \beta - \gamma) + \vec{j}(\alpha + \gamma) + \vec{k}(3\beta - \gamma) \Rightarrow$$

$$\begin{cases} 2\alpha - \beta - \gamma = 2 \\ \alpha + \gamma = 1 \\ 3\beta - \gamma = 1 \end{cases} \Rightarrow [A|B] = \left[\begin{array}{ccc|c} 2 & -1 & -1 & 2 \\ 1 & 0 & 1 & 1 \\ 0 & 3 & -1 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 2 & -1 & -1 & 2 \\ 0 & 3 & -1 & 1 \end{array}\right] \overset{L_2 - 2L_1}{\sim} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & -1 & -3 & 0 \\ 0 & 3 & -1 & 1 \end{array}\right] \overset{L_3 + 3L_2}{\sim}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & -1 & -3 & 0 \\ 0 & 0 & -10 & 1 \end{array}\right] \Rightarrow \begin{cases} \alpha + \gamma = 1 \\ -\beta - 3\gamma = 0 \\ \gamma = -\frac{1}{10}, \ \alpha = \frac{11}{10}, \ \beta = \frac{3}{10} \end{cases}$$

$$\Rightarrow \vec{d} = \frac{11\vec{a} + 3\vec{b} - \vec{c}}{10}$$

1.12) $A(1, -1, 2)$, $B(2, 1, 0)$, $C(3, 2, -6)$

a) $\vec{AB} = \vec{i} + 2\vec{j} - 2\vec{k}$

$|\vec{AB}| = \sqrt{1 + 4 + 4} = 3$

b) $\vec{BC} = \vec{i} + \vec{j} - 6\vec{k}$ $\quad \vec{AC} = 2\vec{i} + 3\vec{j} - 8\vec{k}$

$\vec{AB} \cdot \vec{BC} = 1 + 2 + 12 = 15$

c) $A_{\triangle ABC} = \frac{|\vec{AB} \times \vec{AC}|}{2} = \frac{\sqrt{117}}{2}$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & -2 \\ 2 & 3 & -8 \end{vmatrix} = -10\vec{i} + 4\vec{j} - \vec{k}$$

$|\vec{AB} \times \vec{AC}| = \sqrt{100 + 16 + 1} = \sqrt{117}$

1.13) $\vec{u} = 2\vec{i} + 3\vec{j} + 2\vec{k}$
$\vec{v} = -\vec{i} + 4\vec{j} + \vec{k}$
$\vec{w} = -2\vec{j} + 2\vec{k}$

a) $\vec{v} + \vec{w} = -\vec{i} + 2\vec{j} + 2\vec{k}$

b) $2\vec{u} - 3\vec{w} = 10\vec{i} + 6\vec{j} - \vec{k}$

c) $\vec{u} \cdot \vec{v} = -2 + 12 + 2 = 12$

d) $|\vec{v}| = \sqrt{1 + 16 + 1} = 3\sqrt{2}$

e) $$\vec{u} \times \vec{w} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 2 \\ 0 & -2 & 1 \end{vmatrix} = 7\vec{i} - 2\vec{j} - 4\vec{k}$$

f) $$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 2 \\ -1 & 4 & 0 \end{vmatrix} = -8\vec{i} - 2\vec{j} + 11\vec{k}$$

$|\vec{u} \times \vec{v}| = \sqrt{64 + 4 + 121} = \sqrt{189}$

g) $h = \frac{|\vec{u} \times \vec{v}|}{|\vec{v}|} = \frac{\sqrt{189}}{\sqrt{18}}$

h) $V = \left|\begin{matrix} 2 & 3 & 2 \\ -1 & 4 & 1 \\ 0 & -2 & 2 \end{matrix}\right| \cdot \frac{1}{6} = (20 + 6 + 4) \cdot \frac{1}{6} = 5$

1.14) $A(0,1,-1)$, $B(1,1,3)$, $C(2,1,-2)$, $D(3,-1,2)$

a) $\vec{AB} = (1,0,4)$
$\vec{AC} = (2,0,-1)$
$\vec{AD} = (3,-2,3)$

$V_{ABCD} = \frac{1}{6}\left|\begin{matrix} 1 & 0 & 4 \\ 2 & 0 & -1 \\ 3 & -2 & 3 \end{matrix}\right| = \frac{1}{6}|(-2-16)| = \frac{18}{6} = 3$

b) $V_{ABCD} = d(D, ABC) \cdot A_{ABC}$

$A_{ABC} = |\vec{AB} \times \vec{AC}| = 9$

$\vec{AB} \times \vec{AC} = \left|\begin{matrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 4 \\ 2 & 0 & -1 \end{matrix}\right| = 9\vec{j}$

$d(D, ABC) = \frac{3V_{ABCD}}{A_{ABC}} = \frac{18}{9} = 2$

c) $\sphericalangle(\vec{AB}, \vec{CD}) = \alpha$
$\vec{CD} = \vec{i} - 2\vec{j} + 4\vec{k}$

$|\vec{AB}| = \sqrt{17}$
$|\vec{CD}| = \sqrt{21}$

$\vec{AB} \cdot \vec{CD} = 1 + 16 = 17$

$\Rightarrow \cos\alpha = \frac{\vec{AB} \cdot \vec{CD}}{|\vec{AB}| \cdot |\vec{CD}|} = \frac{\sqrt{17}}{\sqrt{21}} \Rightarrow \alpha = \arccos\left(\frac{\sqrt{17}}{\sqrt{21}}\right)$

1.15) $\vec{a} = 2\vec{i} + 3\vec{j} + (2\lambda - 3)\vec{k}$

$\vec{b} = (2 + 3\lambda)\vec{i} + \lambda\vec{j} - 2\vec{k}$

$\vec{a} \perp \vec{b} \iff a \cdot b = 0 \Rightarrow 4 + 6\lambda - 4\lambda + 6 = 0$

$2\lambda = -10 \Rightarrow \lambda = -5$

1.16) $|\vec{a}| = 4 \quad (3\vec{a} + 5\vec{b}) \perp (\vec{a} - 2\vec{b})$

$|\vec{b}| = 2$

$(3\vec{a} + 5\vec{b}) \cdot (\vec{a} - 2\vec{b}) = 0$

$\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos\alpha = 8\cos\alpha$

$3|\vec{a}|^2 - 6\vec{a}\cdot\vec{b} + 5\vec{a}\cdot\vec{b} - 10|\vec{b}|^2 = 0$

$48 - \vec{a}\cdot\vec{b} - 40 = 0$

$8 = \vec{a}\cdot\vec{b} \iff 8\cos\alpha = 8 \Rightarrow \alpha = 0$

1.17) $\vec{a} = 6\vec{i} + 2\vec{j} - 3\vec{k}$

$\vec{b} = -3\vec{i} + 6\vec{j} - 2\vec{k}$

$\vec{a} \perp \vec{b} \iff \vec{a}\cdot\vec{b} = 0 \Rightarrow -18 + 12 + 6 = 0 \iff 0 = 0$

$|\vec{a}| = \sqrt{36 + 4 + 9} = 7$

$|\vec{b}| = \sqrt{9 + 36 + 4} = 7$ $\Rightarrow L = |\vec{a}| = |\vec{b}| = 7$

$\vec{c} = \pm \dfrac{(\vec{a} \times \vec{b})}{L} \qquad \vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 6 & 2 & -3 \\ -3 & 6 & -2 \end{vmatrix}$

$= 14\vec{i} + 21\vec{j} + 42\vec{k} \Rightarrow \vec{c} = \pm(2\vec{i} + 3\vec{j} + 6\vec{j})$

1.18) $\vec{a} = (2, 3, 1) \qquad \vec{c} = (-2, 1, 2)$

$\vec{b} = (1, 1, -2)$

a) $\vec{a} \cdot (\vec{b} + \vec{c}) = -2 + 6 + 0 = 4$

$\vec{b} + \vec{c} = (-1, 2, 0)$

b) $\vec{a} \times (\vec{b} + \vec{c}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 1 \\ -1 & 2 & 0 \end{vmatrix} = -2\vec{i} - \vec{j} + 7\vec{k}$

c) $\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} 2 & 3 & 1 \\ 1 & 1 & -2 \\ -2 & 1 & 2 \end{vmatrix} = 8 + 6 + 3 = 17$

d) $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{a} \cdot \vec{c}) - \vec{c} (\vec{a} \cdot \vec{b})$

$\vec{a} \cdot \vec{c} = -4 + 3 + 2 = 1$

$\vec{a} \cdot \vec{b} = 2 + 3 - 2 = 3$

$\Rightarrow \vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} - 3\vec{c} = (7, -2, -8)$

1. b) $\vec{DA} \cdot \vec{BC} + \vec{DB} \cdot \vec{CA} + \vec{DC} \cdot \vec{AB} = 0$

$\vec{DA} = \vec{a} - \vec{d}$ $\quad \vec{DB} = \vec{b} - \vec{d}$ $\quad \vec{DC} = \vec{c} - \vec{d}$

$\vec{BC} = \vec{c} - \vec{b}$ $\quad \vec{CA} = \vec{a} - \vec{c}$ $\quad \vec{AB} = \vec{b} - \vec{a}$

$\vec{DA} \cdot \vec{BC} = \vec{ac} - \vec{ab} - \vec{dc} + \vec{bd}$

$\vec{DB} \cdot \vec{CA} = \vec{ab} - \vec{bc} - \vec{ad} + \vec{dc}$

$\vec{DC} \cdot \vec{AB} = \vec{bc} - \vec{ac} - \vec{db} + \vec{ad}$

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$= \vec{ac} - \vec{ad} - \vec{dc} + \vec{bd} + \vec{ab} - \vec{bc} - \vec{ad} + \vec{dc} + \vec{bc} - \vec{ac} - \vec{db} + \vec{ad}$

$= 0$

b) if $\overrightarrow{DA} \perp \overrightarrow{BC} \Rightarrow \overrightarrow{OA} \cdot \overrightarrow{BC}$ (1)

if $\overrightarrow{DB} \perp \overrightarrow{CA} \Rightarrow \overrightarrow{DB} \cdot \overrightarrow{CA}$ (2)

$\overrightarrow{DA} \cdot \overrightarrow{BC} + \overrightarrow{DB} \cdot \overrightarrow{CA} + \overrightarrow{DC} \cdot \overrightarrow{AB} = 0$ (3) $\Rightarrow \overrightarrow{DC} \cdot \overrightarrow{AB} = 0$

$\Rightarrow \overrightarrow{DC} \perp \overrightarrow{AB}$

1.20) $\vec{g} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$

a) $\overrightarrow{AG} + \overrightarrow{BG} + \overrightarrow{CG} = \vec{0}$

$\vec{g} - \vec{a} + \vec{g} - \vec{b} + \vec{g} - \vec{c} = \vec{0}$

$3\vec{g} - (\vec{a} + \vec{b} + \vec{c}) = \vec{0}$

$\vec{a} + \vec{b} + \vec{c} - (\vec{a} + \vec{b} + \vec{c}) = \vec{0} \Rightarrow \vec{0} = \vec{0}$

b) $3\overrightarrow{MG} = \overrightarrow{MA} + \overrightarrow{MB} + \overrightarrow{MC}$

$3\vec{g} - 3\vec{m} = \vec{a} + \vec{b} + \vec{c} - 3\vec{m}$

$\vec{0} = \vec{0}$

1.21) $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 \cdot |\vec{b}|^2$

(Lagrange identity)

$|\vec{a}|^2 \cdot |\vec{b}|^2 \cdot \sin^2\theta + |\vec{a}|^2 \cdot |\vec{b}|^2 \cdot \cos^2\theta = |\vec{a}|^2 \cdot |\vec{b}|^2$

$\Leftrightarrow |\vec{a}|^2 \cdot |\vec{b}|^2 (\sin^2\theta + \cos^2\theta) = |\vec{a}|^2 |\vec{b}|^2$

$\Leftrightarrow |\vec{a}|^2 \cdot |\vec{b}|^2 = |\vec{a}|^2 \cdot |\vec{b}|^2$ (A)

1.22) $|\vec{w}| = 3$ $\quad W = (x, y, 0)$

$\vec{w} \perp Oz$ $\quad Ox(1, 0, 0)$

$\measuredangle(Ox, \vec{w}) = 45^\circ$

$Ox \cdot w = X \Rightarrow Ox \cdot w = |w| \cdot |Ox| \cdot \cos 45^\circ$

$x = 3 \cdot \cos 45^\circ = \frac{3\sqrt{2}}{2}$

$|\vec{w}| = \sqrt{\frac{18}{4} + y^2} \Rightarrow 9 = \frac{9}{2} + y^2 \Rightarrow y^2 = \frac{9}{2} \Rightarrow y = \pm \frac{3\sqrt{2}}{2}$

$\Rightarrow \vec{w} = \frac{3\sqrt{2}}{2}(\vec{i} \pm \vec{j})$

1.23) $\vec{v} = -\frac{\sqrt{3}}{2}\vec{i} + \frac{1}{2}\vec{j}$ $\quad \vec{v} \cdot Ox = -\frac{\sqrt{3}}{2}$

$Ox = (1, 0, 0)$ $\quad \cos\alpha = \frac{\vec{v} \cdot Ox}{|\vec{v}| \cdot |Ox|} = \frac{-\frac{\sqrt{3}}{2}}{1} = -\frac{\sqrt{3}}{2}$

$|\vec{v}| = \sqrt{\frac{3}{4} + \frac{1}{4}} = 1$ $\quad \Rightarrow \alpha = \frac{5\pi}{6}$

1.24) $\vec{v} = 3\vec{i} - \vec{j} + 2\vec{k}$

$\vec{u} = \vec{j} - 2\vec{k}$

$$\vec{v} \times \vec{u} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -1 & 2 \\ 0 & 1 & -2 \end{vmatrix} = 0\vec{i} + 6\vec{j} + 3\vec{k}$$

$|\vec{v} \times \vec{u}| = \sqrt{36 + 9} = \sqrt{45} = 3\sqrt{5}$

$\Rightarrow |\vec{v}| = \sqrt{9 + 1 + 4} = \sqrt{14}$

$\Rightarrow h = \frac{|\vec{v} \times \vec{u}|}{|\vec{v}|} = \sqrt{\frac{45}{14}}$

1.25) $\vec{a} = 3\vec{i} - \vec{j} + 2\vec{k}$

$\vec{b} = \vec{j} - 2\vec{k}$

$\vec{c} = \vec{j} + 4\vec{k}$

$V = \vec{a} \cdot (\vec{b} \times \vec{c}) =$

$$\begin{vmatrix} 3 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & 1 & 4 \end{vmatrix} = 18$$

$$\vec{a}\times\vec{b}=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\3&-1&2\\0&1&-2\end{vmatrix}=6\vec{j}+3\vec{k}\Rightarrow|\vec{a}\times\vec{b}|=$$

$$\sqrt{45}\Rightarrow h=\frac{\vec{a}\cdot(\vec{b}\times\vec{c})}{|\vec{a}\times\vec{b}|}=\frac{18}{\sqrt{45}}=\frac{6\sqrt{5}}{5}$$

1.26) $\vec{a}=3\vec{i}-\vec{j}+\alpha\vec{k}$ $\quad \vec{a}\times(\vec{b}\times\vec{c})=$

$\vec{b}=\vec{j}+2\vec{k}$ $\quad \vec{b}\cdot(\vec{a}\cdot\vec{c})-\vec{c}(\vec{a}\cdot\vec{b})$

$\vec{c}=3\vec{i}-\vec{k}$ $\quad \vec{a}\cdot\vec{c}=9-\alpha$

$\vec{a}\cdot\vec{b}=-1+2\alpha$

$\vec{a}\times(\vec{b}\times\vec{c})\parallel yOz$

$\Rightarrow x=0\Rightarrow \vec{i}(-3+6\alpha)=0\Rightarrow\alpha=\frac{1}{2}$

1.27) $\vec{a}=\vec{i}+\vec{j}+2\vec{k}$ $\quad \vec{a}\times(\vec{b}\times\vec{c})=\vec{b}\cdot(\vec{a}\cdot\vec{c})-$

$\vec{b}=2\vec{i}-\vec{j}+\lambda\vec{k}$ $\quad \vec{c}(\vec{a}\cdot\vec{b})$

$\vec{c}=\vec{i}-2\vec{j}+\vec{k}$ $\quad \vec{a}\cdot\vec{c}=1-2+2=1$

$\vec{a}\cdot\vec{b}=2-1+2\lambda=1+2\lambda$

$\vec{a}\times(\vec{b}\times\vec{c})\parallel xOy\Rightarrow z=0$

$\lambda-1-2\lambda=0\Rightarrow\lambda=-1$

1.28) $\vec{a}=\vec{i}-m\vec{j}+3\vec{k}$ $\quad \vec{a},\vec{b},\vec{c}$ coplanar $\Leftrightarrow$

$\vec{b}=m\vec{i}-\vec{j}+\vec{k}$

$\vec{c}=3\vec{i}+\vec{j}-\vec{k}$

$$\begin{vmatrix}1&-m&3\\m&-1&1\\3&1&-1\end{vmatrix}=0$$

$$\Leftrightarrow \begin{matrix} \\ L_2 - mL_1 \\ L_3 - 3L_1 \end{matrix} \left| \begin{matrix} 1 & -m & 3 \\ 0 & m^2-1 & 1-3m \\ 0 & 3m+1 & -10 \end{matrix} \right| = 0 \Leftrightarrow 10 - 10m^2 + 9m^2 - 1 = 0$$

$$\Rightarrow -m^2 + 9 = 0 \Rightarrow m = \pm 3$$

For $m = 1 \Rightarrow V = \left| \begin{matrix} 1 & -1 & 3 \\ 1 & -1 & 1 \\ 3 & 1 & -1 \end{matrix} \right| = |-1+9| = 8$

1.32) $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{c} \times (\vec{a} \times \vec{b}) + \vec{b} \times (\vec{c} \times \vec{a}) = \vec{0}$

(Jacobi's identity)

$$\vec{b} \cdot (\vec{a} \cdot \vec{c}) - \vec{c} \cdot (\vec{a} \cdot \vec{b}) + \vec{a} \cdot (\vec{c} \cdot \vec{b}) - \vec{b} (\vec{c} \cdot \vec{a})$$
$$+ \vec{c} (\vec{b} \cdot \vec{a}) - \vec{a} (\vec{b} \cdot \vec{c}) = 0$$

1.31). $\vec{a} = x\vec{m} + 2\vec{n}$ $|\vec{m}| = 5$ $\sphericalangle(n, m) = \frac{\pi}{2}$

$\vec{b} = x\vec{m} - 3\vec{n}$ $|\vec{n}| = 3$

a) $\vec{d_1} = \vec{a} + \vec{b} = 2\vec{m} - \vec{n}$ $|d_1|^2 = (\vec{a} + \vec{b})(\vec{a} + \vec{b})$

$\vec{d_2} = \vec{a} - \vec{b} = 5\vec{n}$ $= (2\vec{m} - \vec{n})(2\vec{m} - \vec{n})$

$= 4|\vec{m}|^2 - 4\vec{m}\vec{n} + |\vec{n}|^2 = 4 \cdot 25 + 9 = 109 \Rightarrow |d_1| = \sqrt{109}$

$\vec{m} \cdot \vec{n} = 5 \cdot 3 \cdot \cos\frac{\pi}{2} = 0$

$|d_2|^2 = 5\vec{n} \cdot 5\vec{n} = 25 \cdot 9 \Rightarrow |d_2| = 15$

b) $d_1 \cdot d_2 = |d_1| \cdot |d_2| \cdot \cos\theta \Rightarrow \cos\theta = \frac{-45}{15 \cdot \sqrt{109}} = \frac{-3}{\sqrt{109}}$

$d_1 \cdot d_2 = 10\vec{m} \cdot \vec{n} - 5|\vec{n}|^2 = -45 \Rightarrow \theta = \arccos\left(\frac{-3}{\sqrt{109}}\right)$

c) $\vec{a} \times \vec{b} = (\vec{m} + 2\vec{n}) \times (\vec{m} - 3\vec{n}) =$

$-3(\vec{m} \times \vec{n}) + 2(\vec{n} \times \vec{m}) = -5(\vec{m} \times \vec{n})$

$= |\vec{a} \times \vec{b}| = 5|\vec{m} \times \vec{n}| = 5 \cdot 5 \cdot 3 \cdot \sin\frac{\pi}{2} = 15 \cdot 5 = 75$

1.30) $\vec{u} = 3\vec{a} + 2\vec{b}$ $\quad |\vec{a}| = 2$ $\quad \sphericalangle(\vec{a}, \vec{b}) = \frac{\pi}{3}$

$\vec{w} = 2\vec{a} - \vec{b}$ $\quad |\vec{b}| = 3$

a) $\vec{a} \cdot \vec{b} = 2 \cdot 3 \cdot \frac{1}{2} = 3$

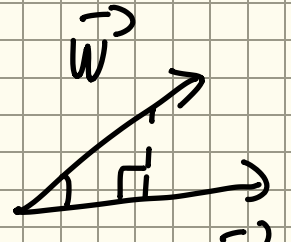
$\vec{u} \cdot \vec{w} = 6|\vec{a}|^2 - 2|\vec{b}|^2 + \vec{a} \cdot \vec{b} = 24 - 18 + 3 = 9$

$|\vec{u}|^2 = 9|\vec{a}|^2 + 4|\vec{b}|^2 + 12\vec{a} \cdot \vec{b} = 36 + 36 + 36 = 108$

$|\vec{w}|^2 = 4|\vec{a}|^2 + |\vec{b}|^2 - 4\vec{a} \cdot \vec{b} = 16 + 9 - 12 = 13$

$\Rightarrow |\vec{u}| = \sqrt{108}$, $|\vec{v}| = \sqrt{13}$ $\Rightarrow \cos\theta = \frac{9}{\sqrt{108} \cdot \sqrt{13}}$

b)



$\cos\alpha = \frac{|\vec{v}|}{|\vec{w}|} \Rightarrow |\vec{v}| = |\vec{w}| \cdot \cos\alpha$

$\Leftrightarrow |\vec{v}| = \frac{|\vec{u} \cdot \vec{w}|}{|\vec{u}|} = \frac{9}{6\sqrt{3}} = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2}$

c) $|\vec{u} \times \vec{v}| = |(3\vec{a} + 2\vec{b}) \times (2\vec{a} - \vec{b})| \ldots$

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1.29) For pos $\vec{a}, \vec{b}, \vec{c}, \vec{h}$ the altitude coord are $(\vec{h}-\vec{a})\cdot(\vec{c}-\vec{b})=0$ and $(\vec{h}-\vec{b})\cdot(\vec{a}-\vec{c})=0$

$AH \perp BC = 0 \Rightarrow (\vec{h}-\vec{a})\cdot(\vec{c}-\vec{b})=0$

$BH \perp AC \Rightarrow (\vec{h}-\vec{b})\cdot(\vec{a}-\vec{c})=\vec{0}$

Se adună și se rezolvă